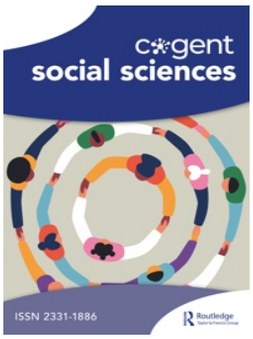

Cogent Social Sciences

ISSN: 2331-1886 (Online) Journal homepage: www.tandfonline.com/journals/oass20

Circular economy and food waste problems in Indonesia: Lessons from the policies of leading Countries

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To cite this article: Waluyo & Dona Budi Kharisma (2023) Circular economy and food waste problems in Indonesia: Lessons from the policies of leading Countries, Cogent Social Sciences, 9:1, 2202938, DOI: 10.1080/23311886.2023.2202938

To link to this article: https://doi.org/10.1080/23311886.2023.2202938

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Waluyo & Kharisma, *Cogent Social Sciences* (2023), 9: 2202938
https://doi.org/10.1080/23311886.2023.2202938

cogent social sciences

Received: 14 February 2023
Accepted: 10 April 2023

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LAW | RESEARCH ARTICLE

Circular economy and food waste problems in Indonesia: Lessons from the policies of leading Countries

Waluyo[1]* and Dona Budi Kharisma[1]

Abstract: Indonesia is the second largest food waste disposer after Saudi Arabia. The potential increase in waste and food waste is an essential issue for Indonesia, given the problem's economic, social, and environmental implications. However, the implementation of the circular economy concept still faces various problems compared to Germany, Netherlands, and China.This paper aims to formulate an ideal policy to implement the circular economy concept through policy comparisons between countries especially to overcome the problem of food waste in Indonesia.

Waluyo

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Waluyo is a lecturer in the Department of Administrative Law, Faculty of Law, Universitas Sebelas Maret Indonesia. In the last five years the author has been concerned with environmental law. Several research titles that have been successfully published in international journals include: The Conservation Of Marine Ecosystem From Trawl Usage By The Local Government Based On Sustainable Marine Preservation Principle, Environmental Protection Pattern Based On Regional Autonomy As The Efforts To Realize Sustainable Development Goals, The Construction of Law System in the Field of Environmental Governance in Realizing Justice and Green Legislation in Indonesia, Social Justice-Based Reconstruction On The Ideal Use Of Coastal And Marine Resources.

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PUBLIC INTEREST STATEMENT

Indonesia is the second largest food waste disposer after Saudi Arabia. The potential increase in waste and food waste is an essential issue for Indonesia, given the economic, social, and environmental costs of this problem. This research examines regulations and policies related to the circular economy in Indonesia, Germany, the Netherlands, and China. The results of this study are helpful as recommendations for any country that will develop regulations related to the circular economy to address food waste and environmental issues. The results of this study are expected to improve the resolution of waste and environmental problems and accelerate the application of the circular economy concept to achieve Sustainable Development Goals (SDGs).

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This is legal politics research that takes a conceptual and comparative approach. A comparative study was conducted on circular economy policies relating to food waste in Indonesia, Germany, the Netherlands, and China. Data was collected through a literature study analyzing various circular economy regulations. According to the research findings, the Circular Economy Act has to be implemented in Germany, Netherlands, and China has been proven effective as a guideline and policy direction to provide solutions for the problems of waste and food waste. The Circular Economy Act serves as a guide and policy direction for the government, private sector, communities, and parties to address environmental issues and help realize sustainable development goals.

Subjects: Asian Law; Environmental Law - Law; Socio-Legal Studies

Keywords: Circular economy; environmental issue; food waste problems; Circular Economy Act; environmental protection

1. Introduction

The climate crisis has caused the earth to experience the hottest temperatures in history, extreme global warming problems, and other climate change impacts that have changed the earth's condition (IPCC, 2012). Moreover, regarding the unsustainable use of natural resources, more than 90 billion tonnes of primary natural resources are mined annually (International Resource Panel, 2019). This puts great pressure on natural resources and ecosystems. There is no denying that humanity is currently on the verge of a sixth mass extinction due to humanity itself (Cowie et al., 2022).

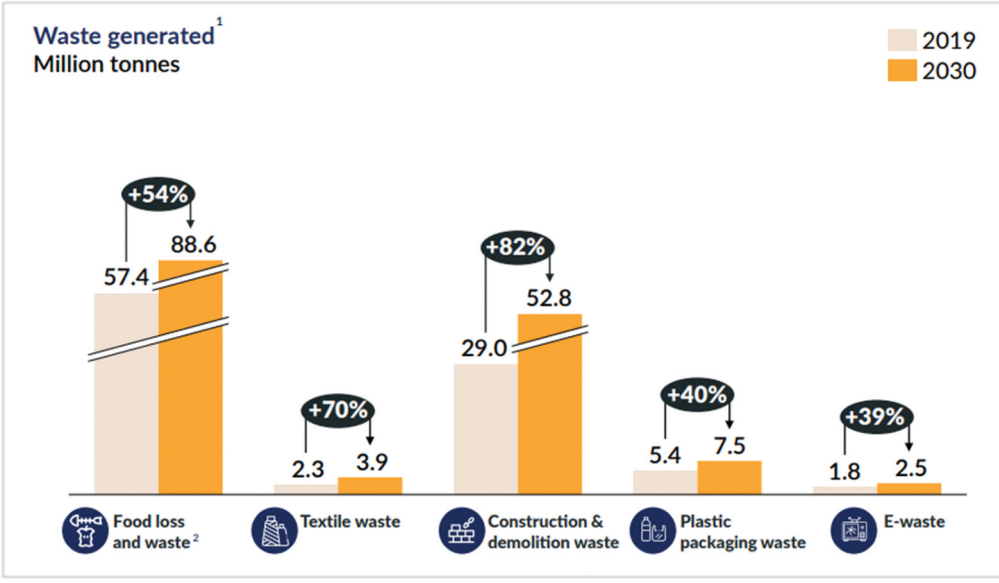
In addition to destroying ecological order and countless species, unsustainable resource extraction and disposal also lead to pollution and a rapidly growing waste problem. Indonesia is no exception to this problem. The research indicates that Indonesia experiences a 54% increase in food loss and waste between the period of 2019 to 2030 (Figure 1) (Bappenas, 2021). Approximately 57 million tons of food was squandered all over Indonesia in 2019, excluding food lost during agricultural production (Ministry of Environment and Forestry, 2017).

Given the economic, social, and environmental repercussions, Indonesia's potential increase in food waste and waste is a major concern (Bappenas, 2021). The Indonesian government spends around USD 1.5 billion on subsidized food for the poor and more than USD 2.3 billion on annual subsidized fertilizers (OECD, 2020b). If Indonesia pledges to reduce food loss and waste, meeting the government's food needs and demands can be accomplished with a lower budget. Additional funds can be directed to other urgent areas such as infrastructure and poverty alleviation.

Moreover, the issue of food waste is closely related to the Sustainable Development Goals (hereinafter referred to as SDGs), specifically SDG 1 and SDG 12. SDG 1 aims to eradicate poverty and reduce inequality, while SDG 12 focuses on promoting responsible consumption and production patterns. Food waste exacerbates poverty by causing the loss of valuable resources that could be used to produce more food, thereby contributing to food insecurity. Additionally, food waste is a significant contributor to unsustainable consumption and production patterns, leading to environmental pollution and negative impacts on biodiversity. Addressing food waste can help create more sustainable consumption and production patterns, contributing to poverty reduction, promoting food security, and protecting the environment, all critical elements of sustainable development.

Figure 1. The non-recyclable waste rate in Indonesia.

Note: Source: Bappenas (2021)



Reducing food waste and waste generation also contributes to reducing food prices for consumers. Consumers in Indonesia pay more than double the price of rice compared to other consumers in Association of Southeast Asian Nations (ASEAN) (Arifin et al., 2018). Furthermore, Indonesia's growing proportion of food imports contributes directly to the country's ongoing deficit (Bappenas, 2021).

In addition, the Indonesian government still faces many obstacles in organizing the waste management system (Hendra, 2016). Indonesia is presently the world's second greatest generator of waste (Hendra, 2016). As a result, it is important to sustainably protect the environment and ensure a secure future for future generations. We need to change how our economy works, how we use materials, and how we dispose of waste.

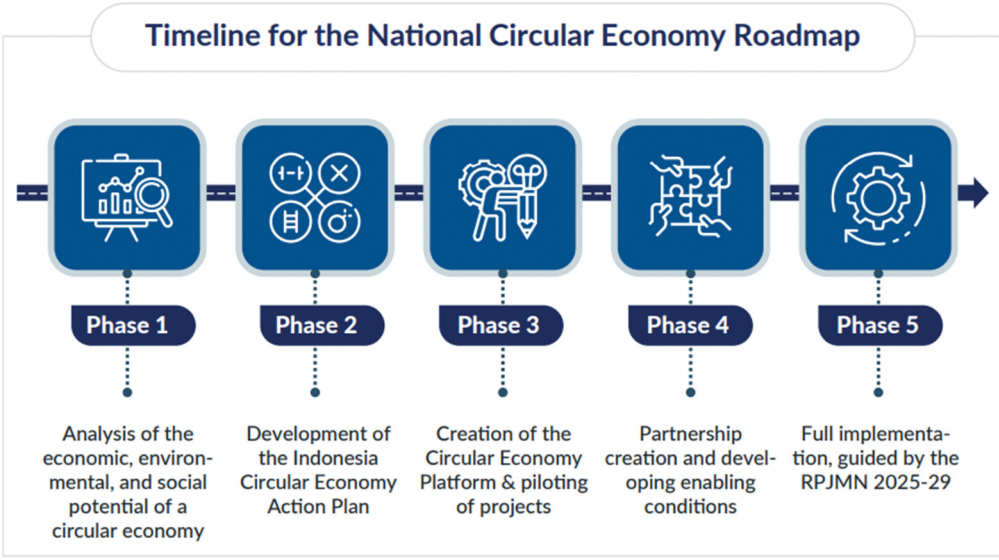
In line with this, the circular economy is considered to be able to answer the related problems. Circular economy refers to an economic concept with a systems-focused approach. It includes restorative or regenerative industrial processes and economic activities that attempt to eliminate waste through materials, design, superior products, and systems by allowing the resources utilized in such processes and activities to keep their worth for as long as possible (including business models) (European Parliament, 2021).

The significance of understanding the impact of the circular economy on reducing food waste, particularly in countries with strong legal protection, cannot be overstated. The circular economy offers a comprehensive and environmentally responsible solution to food waste. Contrasting linear economics, which extracts, uses, and discards resources, the circular economy operates within a closed loop system that minimizes waste and revitalizes resources. Food waste can be reduced in the food business by following circular economy principles such as reducing loss and waste during manufacturing and distribution, reclaiming surplus food, and composting organic waste for new resources.

The circular economy also addresses the root causes of food waste by promoting collaboration among all parties involved in the food supply chain. This collaboration leads to mutually beneficial solutions, benefiting farmers, food producers, retailers, and consumers alike.

Figure 2. Circular economy roadmap timeline in Indonesia.

Note: Source: Bappenas (2021)



In conclusion, the circular economy offers a comprehensive and sustainable solution to the problem of food waste, promoting resource efficiency and waste reduction in the food industry. This system is vital to implement in the Indonesian economy, first and foremost to aim the food waste problem that leads to climate problems.

Currently, circular economy observers and actors are persuading the government to immediately standardize the application of the circular economy idea evenly in Indonesia (Zuraida, 2021). Although the government has prepared a timeline for the adoption of the circular economy based on the provisions of the RPJMN 2025–2029 (Figure 2), not all parties can feel the urgency of the urgency of implementing the circular economy, so the solution to this is a tight regulation with clear boundaries to initiate change.

In general, countries with the highest success rate in the implementation of the circular economy, such as European Union countries such as Germany and the Netherlands, each have a circular economy Act to regulate its implementation; these countries score first and have strong recycling systems in place, as well as high levels of innovation in the circular economy sector (Hervey, 2018; Isle, 2021).

More developed countries also tend to score higher in circular economy implementation because they have more developed economies with investment and patenting arrangements (Hanemaaijer et al., 2021) In the circular economy implementation since they have more developed economies with higher private investment and patenting arrangements (Isle, 2021). Furthermore, European Union member states have sound recycling systems, with Germany's recycling rate touching 66% and the Netherlands' 53% making these countries among the top five countries with the lowest food waste scores (Hervey, 2018).

Considering the growth of the circular economy implementation in both countries, several studies discuss the adoption of the circular economy. First, the research discusses the importance of laws and regulations in overseeing circular economy implementation development (DeLorenzo et al., 2019; Li & Lin, 2016). Second, research conducted by (Zhao, 2020), examine the development of circular economy laws and policies and their implementation at the local level, from industrial areas to cities including research for implementation. Third, a study discusses the

evaluation of the progress of the Dutch Government's goal to accomplish a fully circular economy by 2050 and the progress of Dutch policies regarding the adoption of a circular economy (Hanemaaijer et al., 2021).

However, according to the literature review, no research addresses the issue of food waste and the implementation of a circular economy, particularly in Indonesia. Besides, a comparative analysis of the urgency of establishing and implementing a circular economy act to address the national waste problem must be researched and studied since the act is critical to the successful implementation of the circular economy concept. They provide a framework and standards for implementation, coordinate the transition, and encourage investment and innovation. Policies also regulate resource efficiency and waste reduction, address market failures, and provide support for circular initiatives.

Therefore, it is important to understand the extent to which the provisions regarding the circular economy's implementation in reducing waste, particularly food waste, are applied in countries with a strong legal framework and protection. In this regard, the researcher examines the circular economy policy and regulation in Indonesia to compare with circular economy policy and regulation in European Union countries (Germany and the Netherlands) and China.

This paper seeks to explore two questions which include: how does Indonesia's circular economy policy compares to Germany, Netherlands, and China, which have successfully implemented the Circular Economy System, and what is the ideal legal construction to implement the circular economy in Indonesia?

2. Methodology

This is a research of legal politics that takes a conceptual and comparative approach. The conceptual approach is used when the researcher must build a concept to be used as a reference in his research without departing from existing legal rules (Marzuki, 2005). It is done because there is no legal rule for the problem.

The comparative approach compares a country's law with another country's law or a particular time's law with another time's law. This research compares circular economy policies in Indonesia, China, and European Union (EU) countries (Germany, Netherlands). Indonesia was chosen as the object of research because Indonesia is currently in the development and adoption stages of the circular economy. The European Union is a forerunner in circular economy adoption and policy (European Parliament, 2021; Hervey, 2018; Langen, 2021), with the Netherlands and Germany as countries that can overcome waste problems with recycling rates above 50% (Hervey, 2018). China is a country in Asia with similar waste problems, but it has comprehensive regulations and has successfully implemented a circular economy (Keller & Zhang, 2017; Zhao, 2020).

The legal approach examines all laws and regulations that relate to the issue under discussion (Marzuki, 2019). This research examines several statutes and regulations controlling circular economy policies in order to acquire legal arguments on the themes under discussion. It involves reviewing and analyzing existing laws, regulations, policies, and other legal instruments that relate to the research topic.

The type of data used in this research is legal material consisting of laws and regulations applicable in Indonesia, Germany, the Netherlands, and China. The primary legal material examined in this study are as follows:

Indonesia

(a) Indonesia Waste Management Act 2008,
(b) Environmental Protection & Management Act 2009
(c) Indonesia Job Creation Act 2020,
(d) Government Regulation Number 81 of 2012 concerning the Management of Household Waste & Similar Household Solid Waste, and
(e) Government Regulation Number 27 of 2020 concerning Specific Waste Management,
(f) Government regulation Number 22 of 2021 concerning the Implementation of Environmental Protection and Management,
(g) Presidential Regulation Number 97 of 2017 on National Policy and Strategy for Household Waste Management and Waste Similar to Household Waste,
(h) Minister of Agriculture Regulation Number 44 of 2000 also details the rules regarding Good Handling Practices (GHP) to reduce post-harvest food losses,
(i) Minister of Environment Regulation Number 13 of 2012 concerning Guidelines for the Implementation of Reduce, Reuse, and Recycle through Waste Banks,
(j) Minister of Environment Regulation Number 6 of 2022 concerning the National Waste Management Information System.

Germany

(a) Closed Cycle Management and Waste Act 2012
(b) Circular Economy Act (Kreislaufwirtschaftsgesetz) 2012
(c) Circular Economy Action Plan

Netherlands

(a) EU Circular Economy Act
(b) The National Agreement on Circular Economy by 2050 (2016)
(c) The Dutch Raw materials agreement 2016
(d) Circular Economy Action Plan 2016

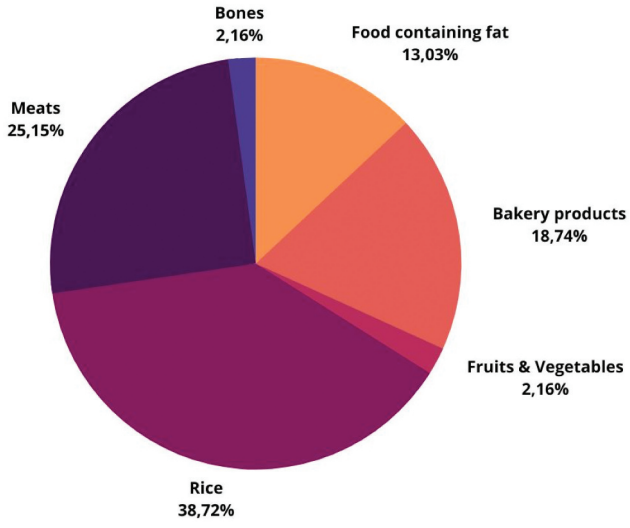
China

(a) China Circular Economy Promotion Law (2008)
(b) Interim Method for Auditing Cleaner Production Projects (2004)
(c) Preferential Income Tax Lists of the Enterprises for Producing Special Safety Production Devices (2008)
(d) Notice on Opinions of Supporting Policies and Measures of Circular Economic Development Investment and Financing (2010)
(e) Interim Management Method of the Central Financial Rewarding Fund for Energy Saving Technologies (2011)
(f) Interim Management Method Circular Economy Development Funds (2012)
(g) Catalog of Encouraged Technologies, Processes, and Equipment of Circular Economy (the First Batch) (2012)

This paper employs a case study technique to developing theories that explain how specific factors or variables interact to produce certain outcomes (Yin, 2018). The technique of obtaining legal materials was carried out by literature study techniques (Kharisma & Hunaifa, 2022; Sudarwanto et

Figure 3. Composition of Food Waste in Indonesia.

Note: Source: Mulyadi (2019)



al., 2021). The literature research method was used to read, examine, review, and take notes on books, laws, rules, papers, and publications about the circular economy.

3. Results and discussion

3.1. Food waste problem in indonesia

Food waste is an ongoing global problem. Every year, one-third of all food grown for human consumption in the world ends up as waste, with a figure of 1.3 billion tons (per year) (Mulyadi, 2019). Based on UNEP (United Nations Environment Program), 61 percent of food waste comes from households, 26 percent from food services such as restaurants/cafes, and 13 percent from the retail industry (UNEP, 2021).

Food waste can be divided into two categories: 1) Food waste that results from generous serving due to the excessive culture of urban society and is called “leftover,” and 2) Food waste that occurs due to planning and management errors. Both are still suitable for consumption and not for consumption (expired food, production errors, and failed products) (UNEP, 2021; Mulyadi, 2019).

Indonesia is the second largest food waste disposer after Saudi Arabia (Mulyadi, 2019). According to the Indonesian Government’s estimated data, about 57.4 million tons of waste were produced in 2019, which covers food waste as well as waste that is unrelated to food production (Bappenas, 2021). The composition of food waste is shown in Figure 3.

On a weekly scale, Around 10 million pieces of bread, cakes, and pastries are discarded in Indonesia. This works up to 292,000 tons of CO2 emissions, which is the value of Indonesia’s annual CO2 emissions. (Susilo et al., 2021). Food waste is organically generated organic waste that endangers public health and the environment by contaminating natural ecosystems.

In addition, the Government of Indonesia still faces many obstacles in organizing the waste management system (Hendra, 2016). Currently, Indonesia is the world’s second largest producer of garbage (Hendra, 2016). Referring to National Waste Management Information System data, the amount of waste generated in Indonesia each year amounts to more than 29 million tons, with a huge amount of waste coming from food waste (SIPSN, 2021).

Figure 4. Waste Composition Based on Waste Type.

Note: Source: SIPSN (2021)

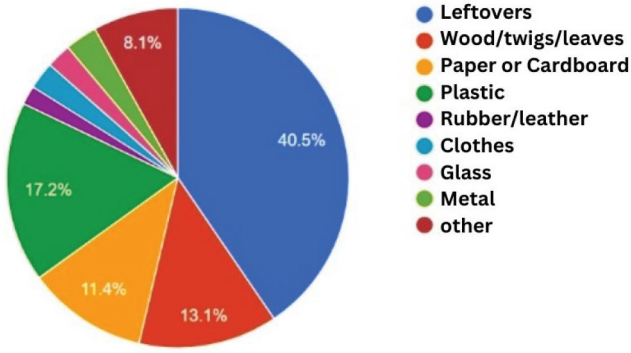


Figure 5. Waste Composition Based on Waste Source.

Note: Source: SIPSN (2021)

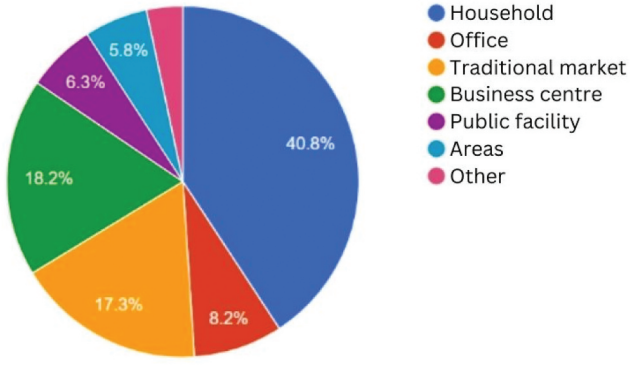


Figure 4 and 5 show the level of waste based on the most significant contributor of waste types and sources. Based on the waste composition, food waste is ranked first in Figure 4, having a yearly value of up to 40.5% of total waste generation. In addition, Figure 5 describes the sources of waste, with households as the most significant contributor. Food waste is a serious problem that costs countries billions of dollars and damages the environment, contributing to global warming and climate change (Ministry of Environment and Forestry, 2017). Food waste ends up in landfills and produces greenhouse gas called methane, which is 28 times more potent than the heat-trapping power of carbon dioxide (Globe, 2021). To date, about 70 percent of the waste entering landfills in Indonesia is organic waste, mainly from food waste (Bappenas, 2021).

3.2. Urgency of circular economy concept to address food waste problem

The circular economy is a production and consumption paradigm that tries to extend the product, raw material, and resource life cycles. In this way, the product's life cycle is extended (European Parliament, 2021).

In reality, the circular economy reduces waste to a minimum. Waste prevention is a high concern in a sustainable system. Products should be used for as long as possible. At the end of the cycle, the waste from the item should be treated as much as possible and sent for disposal or landfill. In this way, it is possible to avoid negative consequences on human health and the natural environment (Van Der Veen & Kapadia, 2021).

The alarming rate of food waste in Indonesia has a significant impact on the environment. About 70 percent of the trash disposed of in landfills in Indonesia is organic waste, mainly from food

waste (Ministry of National Development Planning of the Republic of Indonesia, 2021). Over the past two decades, total Green House Gas (GHG) emissions from food loss/waste (FLW) amounted to 1,702.9 megatonnes of CO2. The amount of FLW generated in Indonesia from 2000 to 2019 was 23–48 million tonnes/year, or 115–184 equivalent to kg/capita/year (P. P. N. Bappenas et al., 2021).

That problem can be reduced if a circular system replaces the usual linear take-make-waste system of dumping waste into landfills. This circular production and consumption in principle reduces the utilization of raw materials and the processing of raw materials' resources (Kharisma, 2020). By switching to this circular system, such materials are much more efficient and minimize the environmental and natural resource pressures associated with resource use (Hanemaaijer et al., 2021).

As a member of the global development agenda, Indonesia is committed to supporting the SGDs and Greenhouse Gas (GHG) reduction targets under the 2030 Paris Agreement. Indonesia is currently transitioning towards this circular system. The Ellen MacArthur Foundation established the circular economy concept 2009, but the idea only reached Indonesia in 2018–2019. Circular economy concepts take ten years to infiltrate Indonesian waste management stakeholders (European Parliament, 2021).

As a way to address the issue of food waste in Indonesia, the circular economy has to be implemented. With a high rate of food waste and a large population putting pressure on limited resources, it becomes imperative to find a systemic solution. The circular economy provides such a solution by promoting a closed-loop system that reduces waste and maximizes resource use.

In addition, the circular economy fosters collaboration between all stakeholders in the food industry, leading to more efficiency and sustainability. This is especially important in Indonesia where the food industry plays an important function in supporting the economy and population. Therefore, it is imperative that the circular economy act be applied urgently in Indonesia to solve the problem of food waste and support the country's sustainable development goals.

3.3. Problems and challenges of circular economy implementation

The issue of waste in Indonesia can be solved by adopting a circular economy. However, the dawning of the circular economy involves a number of issues that must be addressed right now. Some of these challenges are:

3.3.1. The absence of legislation regulating the circular economy

Regulation is critical to the development of the circular economy. In some countries, circular economy implementation is regulated explicitly in one integrated legal instrument document. For example, Germany and the Netherlands ratified the European Union's "Circular Economy Act" and "Integrated Product Policy Directive" as the core guidelines for the application of a circular economy across sectoral policies, and China with its "Circular Economy Promotion Law" for nearly 20 years have successfully transformed the country into a country with a strong circular economy policy, making great strides by regulatory and administrative measures at the federal, state, municipal, industry, and factory levels (Bleischwitz et al., 2022).

Indonesia does not yet have a national legal instrument that explicitly regulates the circular economy integratively. Although several multi-sectoral laws contain provisions regulating sustainable economic activities, in practice, there are still discrepancies in the implementation of these regulations due to the absence of technical regulations regarding the implementation mechanism of the relevant regulations. Therefore, there is a need to draft an integrative circular economy act so that from the legal substance aspect, the circular economy can be optimally regulated and applicable to all sectors to contribute the circular economy implementation in Indonesia.

The readiness of legal instruments plays an important role and is a determining factor in the implementation of the circular economy. The substance of laws and regulations can be an obstacle in several conditions, such as legal uncertainty, overlapping regulations, complexity, and unclear procedures that lead to inefficiency.

Additionally, the legal policy direction in Indonesia related to Circular Economy and Food Waste issues is still evolving. Despite the Indonesian government has issued several policies and programs to address these issues, such as the 3 R (Reduce, Reuse, Recycle) program and regulations concerning trash management. However, the implementation of policies and programs still poses numerous challenges, such as lack of support and coordination between government agencies and businesses.

Therefore, further efforts are still needed to optimize the legal policy direction related to Circular Economy and Food Waste in Indonesia. Every area of regulation must be appropriate and supportive of sustainable economic development and environmental sustainability. Currently, the nomenclature or terminology of "circular economy" has not been regulated in the content material of statutory provisions in Indonesia (Fasa, 2021).

Promoting the circular economy cannot be done by one sector alone. In other words, as it is multi-sector, the process requires synergistic participation of governments, communities and all relevant stakeholders (Hysa et al., 2020). Therefore, an upstream-to-downstream regulatory scheme is essential.

3.3.2. Lack of synergy among stakeholders

Implementing a circular economy requires significant investment and resources, which can be a challenge in a country with limited resources. Policy development stakeholders include government, business/industry, academia, and civil society (Hysa et al., 2020). The government, as the regulator, has not fully facilitated the legal framework, financing mechanism, and governance of the circular economy in Indonesia. In supporting academia in research and innovation of environmentally friendly technologies and waste recommendations, as well as the industry sector as a developer of business models, products, and implementation of sustainable production, the government needs to input these materials specifically in legislation so that the direction and goals of development can be set with the same point of view.

In addition, the Indonesian people's participation is still minimal, considering that people in the household sector primarily produce the protracted waste problem. Many people are still unaware of the benefits of a circular economy and the influence it may have on trash reduction and sustainability. This lack of awareness makes it difficult to generate support for circular economy initiatives. Therefore, information literacy in the community needs to be carried out through policy monitoring and evaluation.

3.3.3. Policies in circular economy financing/budgeting

The government of Indonesia has yet to provide significant support for the execution of circular economy policies, including investment in the necessary infrastructure and institutions. Financing policies are fundamental in the enforcement of a circular economy. It is equally critical that these budgets be made available to local governments, as is the case in China.

Indonesia has not explicitly regulated the circular economy budget in either the Indonesian's State Budget or Regional Budget. This condition impacts the budget allocation for the circular economy, which does not yet exist. Besides, the development and implementation of circular economy policies require a budget; this is both a challenge and a problem faced by stakeholders.

In conclusion, there are several barriers that need to be overcome in order to fully implement circular economy policies in Indonesia. Addressing these challenges will require a coordinated

effort from the government, businesses, and communities, as well as investment in the necessary infrastructure and institutions.

3.3.4. Statistics office

Indonesia does not yet have a specialized statistical office overseeing policy implementation at the sectoral level. Unlike the Netherlands and Germany as European Union member countries that have statistical offices. In practice, this statistical office is vital to support the successful implementation of the circular economy.

Eurostat, the statistical office of the European Union based in Luxembourg, is in charge of accumulating data on the progress of its member states toward set targets. So, in addition to monitoring, it also offers Europe with high-quality statistics.

The statistical office is responsible for two environmental data centers. The center checks whether the various waste streams in EU countries are properly sorted and treated, whether producers are responsible and whether waste policies are effective (Institut, n.d.).

3.4. Country comparison of circular economy implementation regulations and policies

3.4.1. Indonesia

The circular economy in Indonesia is featured under National Priority 1 in the National Medium-Term Development Plan (RPJMN) for 2020–2024. Building economic resilience and achieving national priority 6 for quality and equitable growth: Build the environment to improve resilience to disasters and climate change.

Under National Priority 6, the Circular Economy falls under the umbrella of Low Carbon Development (LCDI), which is also one of the efforts to achieve a green economy by emphasizing

Table 1. Circular Approach Concept in Indonesia

Reduce	(a) eliminating waste in the manufacturing and supply chain (e.g., 3D printing) (b) virtualize products and services (e.g., e-books) (c) reduce energy consumption by increasing energy efficiency (d) redesigning products to use less resources (use of high-strength steel in construction)
Reuse	(a) Sharing assets (cars, motorcycles, electronic equipment) (b) use of used products (c) improve asset utility by providing products as services
Recycle	(a) Recycle materials (b) organic waste biochemical extraction
Refurbish	(a) product or component remanufacturing (b) extend product life through product maintenance
Renew	putting sustainable energy and materials first (replacing plastic packaging with paper-based packaging)

its activities on five priority sectors. Three of the five "Secure, Contain, Protect" (SCP) sectors are closely related circular economy principles, namely waste management, sustainable energy development, and green industry development (LCDI, 2021).

To date, the adoption and implementation of circular economy concepts has not been explicitly regulated by Indonesian laws and regulations. The circular economy concept in Indonesia is implemented with the following approaches (Bappenas, 2021) (See Table 1):

3.4.2. China

In 2004, China's National Development and Reform Commission convened the first national working conference on the circular economy idea to reduce resource consumption and emissions, enhance resource efficiency through the reduction principle, and achieve sustainable growth. Promoted the circular economy as an economic growth framework consistent with sustainable development, reuse, recycle (Zhao, 2020).

In 2005, the State Council issued "Several Opinions on Accelerating Circular Economy Development" (Several Opinions) to organize government efforts from the national to the local levels to implement the circular economy. The government leads China's Circular Economy approach in a top-down, centralized manner.

The central government formulates regulations and country wide laws, goals and standards, and assigns duties and quotas for local enforcement (Zhao, 2020). Currently, China has an environmental law with a system tailored to social situations and long-term economic development, which fundamentally ensures resource efficiency, effective pollution management, and long-term economic development in China (Keller & Zhang, 2017).

The environmental laws of China are gradually playing a more significant and prominent role. The Environmental Protection Law, amended in 2014, called as the most "rigorous" environmental law in history, demonstrating the government's unwavering commitment to boosting the country's long-term development (Keller & Zhang, 2017; Zhao, 2020).

3.4.3. European Union

Resource conservation was recognized as a European priority in 2020. The adopted strategy "Resource-efficient Europe" is explicitly recognized as one of the seven flagship initiatives (European Parliament, 2021). The European Union enacted the Circular Economy in 2018, which includes modifications to core waste management regulations and was implemented into German law through the Circular Economy Act (Institut, n.d.).

In addition, the European Union enacted the Circular Economy in 2018, which includes modifications to core waste management regulations and was implemented into German law through the Circular Economy Act (Hervey, 2018).

The Circular Economy Action Plan suggest legislative measures to promote sustainable product policies to increase the durability of reusable, repairable, and recyclable products. The aim is to maximize product recycling to reduce the dependance on primary raw materials.

3.4.3.1. Germany. Since the German waste management industry has been guided by the principle of "prevention, recovery, then disposal" since the passage of the German Act on the Closed Substance Cycle and Waste Management (Kreislaufwirtschafts- und Abfallgesetz) in 1996. This means that the first priority is to minimize waste generation, followed by comprehensive recycling of any residual energy or materials. Finally, Any remaining garbage must be disposed of in a way that benefits the public (Öko-Institut, 2022.).

Table 2. Comparison of Circular Economy Policy Indicators

Circular Economy Indicators	Indonesia	German	Netherlands	China
Regulation	(a) Indonesia Waste Management Act 2008, (b) Environmental Protection & Management Act 2009, (c) Indonesia Job Creation Act 2020.	(a) Closed Cycle Management and Waste Act (2012), (b) Circular Economy Act (Kreislaufwirtschaftsgeset-z) 2012 (adopted by the Cabinet of the Federation 2020) (c) Circular Economy Action Plan	(a) Adoption of the EU Circular Economy Act (b) The National Agreement on Circular Economy by 2050 (2016) (c) The Dutch Raw materials agreement 2016 (d) Circular Economy Action Plan 2016	(a) China Circular Economy Promotion Law (2008) (b) Interim Method for Auditing Cleaner Production Projects (2004) (c) Preferential Income Tax Lists of the Enterprises for Producing Special Safety Production Devices (2008) (d) Notice on Opinions of Supporting Policies and Measures of Circular Economic Development Investment and Financing (2010) (e) Interim Management Method of the Central Financial Rewarding Fund for EnergySaving Technologies (2011) (f) Interim Management Method Circular Economy Development Funds (2012) (g) Catalog of Encouraged Technologies, Processes, and Equipment of Circular Economy (the First Batch) (2012)

(*Continued*)

Table 2. (*Continued*)

Circular Economy Indicators	Indonesia	German	Netherlands	China
Financing-related policies	Not yet regulated	A flagship partnership through the Kreditanstalt für The KfW Bank in Germany, also known as the Wiederaufbau Bank, aims to invest approximately 10 billion euros from 2018 to 2023 through the IPA national bank. This investment is aimed at reducing and preventing waste, enhancing resource efficiency, and encouraging innovation by promoting circularity across all economic sectors (European Commission, 2018).	(a) It is imperative that financial entities explicitly adopt circularity principles and implement novel elements of circular finance. (b) The integration of circular metrics within financing practices not only increases transparency but also enables external evaluation and comparison, thereby reducing the potential for greenwashing. (c) Provision of early-stage venture capital and the establishment of suitable financing mechanisms for circular businesses can facilitate their growth and development, enabling them to meet the investment criteria of more advanced financiers, including banks and institutional investors. The utilization of blended finance, public funding, and financial innovation can play a pivotal role in this regard.	(a) Projects related to the circular economy should receive supportive credits. (b) Streamlining of loan procedures for projects that encourage investment through the "List of Industry Guidelines to be Adjusted". (c) Prohibiting the extension of credit and cancellation of loans for projects slated for elimination. (d) Forbidding the provision of loans for projects that conflict with policies aimed at supporting and promoting the adjustment of key industries or related industrial policies, particularly those backward projects to be abolished as specified by relevant regulations and laws. (e) Funding for energy conservation and emissions reduction should be allocated by factors such as the property, purpose, investment cost, energy-saving and emissions reduction impact, rate of energy and resource utilization, and other project elements. (f) Subsidized financial support shall be provided. (g) Designated funds should be utilized to support the national urban mineral demonstration base, kitchen waste recycling and safe disposal, circular transformation, and promotion of cleaner production technologies.

(*Continued*)

Table 2. (*Continued*)

Circular Economy Indicators	Indonesia	German	Netherlands	China
Education and Public Awareness	Not yet regulated	(a) The advancement of comprehensive and practical knowledge through its integration into relevant educational programs, including lectures on Circular Economy, Master's degree programs focused on the Circular Economy, and in-depth courses. (b) Raising awareness and educating both the general public and professionals on the fundamental concepts of the Circular Economy, including the conservation of resources and mitigating climate change, with the aim of promoting change in consumer behavior and promoting the adoption of sustainable products, business models, and economic and business practices. (c) The creation and development of specialized job opportunities in the Circular Economy, such as in the field of advanced manufacturing technologies.	Local, regional, national, and worldwide businesses, academic institutions, and non-governmental organizations, with the government serving not only as a market regulator but also, as needed, fulfilling the role of a facilitator, and guiding the direction of the effort. Additionally, the government acts as a network partner, demonstrating its commitment and actively collaborating with various stakeholders.	Increasing public awareness and comprehension of the Circular Economy through increased awareness campaigns and education can effectively drive the adoption of Circular Economy principles in daily life (Qu et al., 2022)

(*Continued*)

Table 2. (*Continued*)

Circular Economy Indicators	Indonesia	German	Netherlands	China
Government Collaboration in Public Procurement on Innovation and Pilot Projects	(a) The Polyethylene Terephthalate (PET) bottle recycling factory is being built in accordance with the government's pledge to supporting sustainable industrial development and the national priority agenda of reducing plastic waste by 70% by 2025. (b) The National Capital City (IKN) develop with a circular concept as the first circular city in Indonesia.	A lack of defined rules for both businesses and the public sector, such as the mandated use of recycled materials in manufacturing, waste reduction goals, and substantial increases in recycling rates. There is a need for more proactive measures to secure a firm public sector commitment to reducing waste and including recycled material criteria into procurement practices (Institut, n.d..).	The implementation of a phased increase in circularity requirements for government purchasing and procurement, including those related to producer responsibility, is required. This approach will ensure that the efficient recycling and reuse of high-quality resources is incorporated into the design of the production process as a standard benchmark.	(a) Cleaner production technologies should be based on expert verification, (b) The government should actively promote the use of mature and advanced cleaner production technologies belum by procuring them.
Monitoring and evaluation	Data monitoring and information by the National Waste Management Information System	The "Municipal Solid Waste Technical Guidelines" have prohibited landfilling without treatment since 2006.	The Cabinet deployed the BIT team on a broader scale and launched a study to gain more ground in influencing norms and values (unconscious behavior) in the circular transition.	Implement compulsory audits of companies that contravene local regulations or national emission standards, utilize toxic substances, or dispose of hazardous materials.

(*Continued*)

Table 2. (*Continued*)

Circular Economy Indicators	Indonesia	German	Netherlands	China
Stakeholder engagement	(a) Issuing Government Regulation 2021 on the Implementation of the Industry Sector as an implementing regulation IJC Act 2020 (b) Guarantee the availability and ease of obtaining raw materials or supporting materials based on natural resources, recycled products, and other policies to increase the competitiveness of business activities and the industrial area.	A "duty of care" (Obhutspflicht) policy that requires distributors in the event of a distance sale to ensure that the goods remain usable if returned by the customer and do not become wasteful.	Ensure a well-defined distribution of responsibilities among stakeholders in the implementation of circular economy policies (Hanemaaijer et al., 2021). There is already a significant amount of research and innovation aligned with the existing system. Of the 1,900 innovative circular companies in the Netherlands, the majority (66%) prioritize recycling.	(a) Income tax benefit for businesses that buy and use specialized equipment for environmental protection, energy efficiency, water conservation, and production safety. (b) Companies utilizing water, gas, and solid waste as primary production materials receive a 5-year income tax reduction or exemption.
Strategies and initiatives	Adopting the circular economy concept in development vision and strategy, especially in 5 (five) industrial sectors, Food and beverages, construction, electronics, textiles, and polymers are just a few examples. The next step is the creation of a National Action Plan and incorporating circular economy as a priority in the 2020–2024 National Medium-Term Development Plan.	The "waste hierarchy" policy states that waste should be avoided first, and waste management planning should also be oriented towards this goal. When this is not possible, waste can be reused, recycled, or (eventually) incinerated.	(a) Develop a more detailed vision of the circular economy that companies and civil society organizations widely support. (b) Circular economy policies should incorporate strong measures such as taxes, regulations, and standardization to be effective. (c) The Dutch recycling industry is a result of decades-long waste management policies that prioritize waste reduction and minimizing landfill use.	Circular economy policies include regulation, tax, fiscal, financial, and pricing measures that aim to improve industrial structure, cleaner production, recycling, waste management, and resource and utilization of energy.

In 2012, the Act underwent a revision with the goal of creating harmonization across the European Union, resulting in the adoption of the Circular Economy Act, which had the same aim. The latest law reform—adopted by the Federal Cabinet on 12 February 2020 - goes a step further and leads the implementation of the circular economy even further.

Despite this, environmental groups have expressed worry about the lack of explicit criteria for enterprises and the public sector, such as mandated recycled material utilization in manufacturing, high waste reduction objectives, and enhanced recycling rates.

Furthermore, Germany and other EU Member States have yet to adopt EU Directives, such as the Packaging Directive, the Waste from Electrical and Electronic Equipment (WEEE/e-waste) Directive, the Batteries Directive, the End-of-Life Vehicles Directive, and the Landfill Directive, into national legislation.

3.4.3.2. Netherlands. The Dutch national government implemented a circular economy strategy in 2016, with the objective of becoming waste-free by 2050. Based on the Netherlands Organization for Applied Scientific Research (TNO), the circular economy has the potential to obtain 7.3 billion euros in related industries and provide employment for up to 54,000 people. Additionally, the strategy could lead to a reduction in raw material usage by 100 megatons, this equates to one-quarter of the Netherlands' annual raw material imports (OECD, 2020a).

The Netherlands is one of the European Union's leaders in circular economy policy. The Netherlands recycles up to 80% of its total waste (Langen, 2021). The Netherlands consumes 20% less raw materials than the rest of the European Union. The EU's circular economy policy plays a crucial role in supporting the Netherlands in its journey towards a waste-free economy. The policy sets standards for using material resources in product design and prohibits the use of harmful substances in products.

This EU policy will promote a more equal market between producers and consumers across its member states, and the Netherlands, with its open economy and advanced waste management policies, will especially reap the benefits. The Netherlands' waste policy has evolved into a circular economy policy over the years, making it well-positioned to take advantage of the EU's policy. (Hanemaaijer et al., 2021).

The Netherlands has placed a strong emphasis on recycling for years when it comes to waste management. In addition, it uses fewer resources for consumption compared to other European nations, resulting in a more efficient utilization of resources within its economy (Hanemaaijer et al., 2021; Langen, 2021). The high population density in the country helps make efficient use of the large amount of raw materials required for infrastructure development.

Being a trailblazer in recycling in Europe, the Netherlands' goals for a circular economy are different from most other European countries. Other countries, such as Italy and Eastern Europe, still have room for improvement by reducing waste in landfills and increasing recycling. However, for the Netherlands, the main challenge is to incorporate high-quality recycling in new products and focus on other circularity approaches such as reuse, design, refurbishment, sharing, and repair (Hanemaaijer et al., 2021).

In 2018, the council was directed to create a plan for a city that is both sustainable and operates within a circular economy by bringing together various initiatives that already exist and focusing on three key areas (Desmond & Asamba, 2019)):

(1) Public procurement: With the aim of encouraging businesses to adopt circular practices, this area focuses on procurement in the service industry and the construction industry.
(2) Waste management: a possibility to re-evaluate waste management and increase recycling, it includes ending contracts with waste companies by 2022 and working towards becoming a waste-free city by 2030.
(3) Knowledge sharing: Creating venues for collaboration across private, public, and non-profit organizations and connecting with knowledge networks.

The Dutch government's waste-free policy is centered on five primary priorities: biomass and food, plastics, the industrial industry, the building sector, and consumer goods. The success of this strategy is supported by proper regulation, financing, and knowledge (Desmond & Asamba, 2019). The government has set aside 300 million euros annually for implementing circular economy initiatives, with this development fund targeted towards addressing various climate-related issues. These funds are also available to local governments (OECD, 2020a).

Furthermore, In the Netherlands the general public understands the importance of a circular economy. One method of raising awareness is through a program that challenges 100 households to adopt a waste-free lifestyle for 100 days (OECD, 2020a).

With the above arrangements in the Netherlands, Germany, and China, the researcher compares circular economy-related policies in Indonesia with similar policies in Germany, the Netherlands, and China to answer the problems and challenges of implementing a circular economy in Indonesia in tables 3 and 4 below.

The table 2 of comparison pertains to the success markers of the circular economy, which encompasses raising awareness, strengthening capacities, collaboration, education, financing, innovation, testing and experimentation, monitoring and assessment, government procurement, regulations, stakeholder involvement, strategy, and programs (OECD, 2021).

From the comparison above (table 3), it can be concluded that Indonesia's regulation is not comprehensive enough to promote the change to a circular economy. It shows that Indonesia does not have any regulations on financing-related policies and public awareness. Furthermore, Indonesia does not have a circular economy act unlike Germany, the Netherlands, and China, which have specific regulations on this matter.

> In terms of financing policies, both German, Netherlands, and China have specific regulations governing the financing of projects related to circular economy or promoting circular economy systems across all sectors of the economy. Financing-related policies are critical to the successful implementation of circular economy activities. A circular economy aims to minimize waste and promote resource efficiency by reusing materials for as long as possible. However, transitioning to a circular economy requiring investments in new technologies and business models. Financing-related policies, such as grants, subsidies, and tax incentives, provide the necessary funding to support circular economy initiatives and incentivize businesses to adopt circular practices. In addition, financial instruments like green bonds and circular economy funds allow investors to finance circular projects and initiatives, promoting the change to a circular economy. Without adequate financing, the adoption of circular economy practices may be challenging, and progress may be slow.
>
> The absence of regulations in education and public awareness is also a hindrance to the successful adoption of a circular economy. Both German, Netherlands, and China have regulations for utilizing both formal and non-formal education as a bridge to increase public awareness to participate in the process of transitioning to a circular economy. When the awareness increase, public are more likely to participate in waste reduction efforts and support initiatives that promote sustainable practices. Moreover, public awareness also influences policymakers to create regulations and policies that promote a circular economy.

When citizens demand sustainable practices and circular economy initiatives, policymakers are more likely to respond by implementing policies and regulations that support these initiatives.

Furthermore, Regulations regarding Government Collaboration in Public Procurement on Innovation and Pilot Projects, and monitoring and evaluation are still insufficient compared to the regulations in Germany, the Netherlands, and China, which focus on prevention by building an environmentally friendly ecosystem starting from the production process. In contrast, Indonesia still focuses on waste management and not on how the waste is produced.

Additionally, Indonesia's regulations are not sufficient to prevent businesses from producing a massive amount of waste. On the other hand, Germany and the Netherlands have regulations on responsibility-sharing and recycling for businesses. Meanwhile, China provides tax incentives, including tax exemptions, for businesses that apply circular system. Indonesian regulations only regulate the ease of obtaining natural resource-based raw materials or auxiliary materials, or recycled products to enhance the competitiveness of business activities and industrial areas. This does not encourage businesses and stakeholders to fully transition to a circular system.

3.5. Ideal policy construction to be applied in the adoption of circular economy policy in Indonesia

From the outcome of the comparison that has been completed, It is visible that the current circular economy-related policies in Indonesia are still lacking and have not targeted the transition. The focus of the Circular Economy transition in Indonesia is still primarily on mitigating the harmful effects of waste and emissions, with limited emphasis on maximizing raw material value.

It is necessary to enact legislation that specifically governs the Circular Economy. Based on comparative analysis with various nations, crucial elements to be regulated within the Circular Economy legislation include the establishment of a regulatory framework for waste management, the promotion of resource efficiency and sustainability, the encouragement of closed-loop production processes, and the imposition of penalties for non-compliance with Circular Economy principles and regulations.:

(a) Standardize waste management inline with the notion of the circular economy;
(b) Increased knowledge and innovation of the parties;
(c) Provision of budget related to financing circular economy projects;
(d) incentive and taxation arrangements;
(e) establishment of a supervisory body and a statistics office;
(f) International cooperation.

The following are some policy recommendations for facilitating the transition to a fully-fledged Circular Economy:

(1) A solid national policy is required to govern the adoption of the circular economy.
(2) Circular economy policies include measures related to taxation, regulation, and standardization. However, it is important to consider that the development and implementation of regulations and legal instruments in this field often requires a considerable amount of time (For example, the plastic excise policy that is being worked on and will be enforced in 2023).
(3) The policy must establish a distinct delineation of responsibilities among the various participants involved in executing circular economy initiatives.
(4) the policy should encompass measures to ensure the secure provision of material resources, promote green tax reform, support sustainable international commerce, foster green

technological advancement, and align educational standards with circular manufacturing methodologies. This will require a coherent approach across ministries where each ministry has its role to play.

(5) Local governments are critical in monitoring the adoption of circular economy regulations, specifically with regards to addressing food waste and other matters pertaining to the adoption of circular economy initiatives at a local level.

(6) The government must take an active role in regulating the waste management industry and the environment at the level of households; as the most significant waste producer each year, it is necessary to increase efforts to measure food and inedible parts that end up being wasted.

The policy should be able to regulate circular economy activities as a whole. The establishment of a new circular economy legal system includes the following aspects. First is establishing a supervisory mechanism that enforces standardized regulations on implementing the circular economy, in this case, to address environmental concerns. The second objective involves instituting stringent environmental quality standards to regulate and mitigate environmental harm.

The third aim is to enact laws to steer the utilization of renewable energy, encourage clean manufacturing and recycling economy, and reduce the utilization of resources. For instance, the legislation mandates the government to offer incentives and China has exhibited favorable policies for the implementation of renewable energy and resource recycling (Keller & Zhang, 2017; Zhao, 2020).

Fourth, improving public knowledge of environmental preservation and sustainable development, especially in promoting and implementing circular economy policies. Finally, Citizens' participation in environmental decision-making laws better incorporate democratic principles and environmental preservation.

4. Conclusions

A circular economy demands a broad and systemic plan that transcends sectoral policies strategies. However, policies in Indonesia have not been able to regulate these matters. Laws and regulations related to environmental management, such as the Environmental Act and the IJC Act 2020 and their derivative regulations, have not been able to oversee the adoption of circular economy policies in Indonesia. The elements of circular economy regulation with a broad and integrated policy focus, ranging from the environment, regional development, agriculture, and circular industry, have not been regulated and accommodated in the regulation. It is crucial to identify these sectors as a first step to avoid implementing disconnected projects due to the absence of a comprehensive approach in the near to medium term.

The scope of this research is restricted to a comparative study conducted on Indonesia in comparison with Germany, the Netherlands, and China in the adoption of circular economy policies to decrease waste and food waste levels using a variety of indicators.

Furthermore, future research interests in the establishment of circular economy act to decrease waste and food waste could include evaluating the effectiveness of circular economy policies and regulations in different contexts and regions, and focus on the design and adoption of circular economy legislation at the local and national levels.

The implications of the research provides insights for industry, government, and society on how the adoption of the circular economy concept can reduce food waste, while also informing the development of more effective business strategies and practices. The research has theoretical implications as well, allowing us understand how the circular economy concept may be utilized in various domains, and can help develop theories on reducing food waste. The act establishing a

circular economy can facilitate the establishment of effective business procedures and policies intended to reduce food waste and promoting the application of circular economy principles in diverse fields

Therefore, there is a need for policy formation, policy coherence, and policy development to create a shared vision across initiatives in the implementation and the transition to a huge circular economy. In this case, it is critical to examine the formation of a circular economy legislation.

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Disclosure statement
No potential conflict of interest was reported by the authors.

Citation information
Cite this article as: Circular economy and food waste problems in Indonesia: Lessons from the policies of leading Countries, Waluyo & Dona Budi Kharisma, *Cogent Social Sciences* (2023), 9: 2202938.

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